DATA STRUCTURE ASSIGNMENT 3

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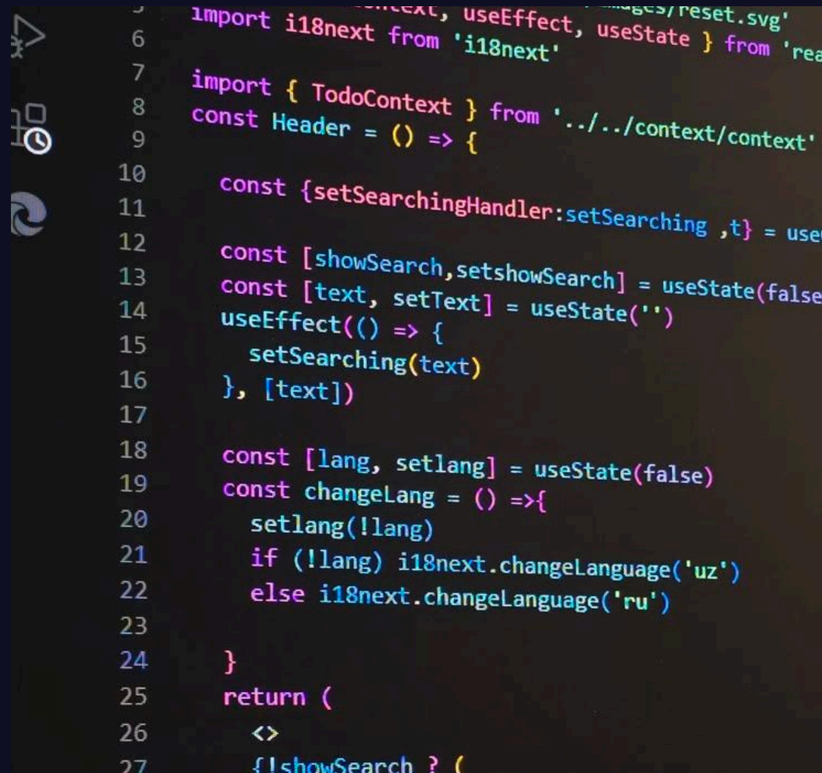
202500283



Q1

Write a C++ program, that takes the course marks for a student, then using functions, decide whether this student passed (his mark is greater than or equal 50) or failed, and display his corresponding grade (A if it's between 100 and 90, B if it's between 90 and 80, and so on, ...).

Source Code

assignment_3 (Global S

```cpp
#include <iostream>
using namespace std;
void pass_or_fail(float mark) {
    if (mark >= 50) {
        cout << "Passed: ";
    }
    else {
        cout << "Failed: ";
    }
    if (mark >= 90) {
        cout << "A";
    }
    else if (mark >= 80) {
        cout << "B";
    }
    else if (mark >= 70) {
        cout << "C";
    }
    else if (mark >= 50) {
        cout << "D";
    }
    else {
        cout << "F";
    }
}
int main() {
    float mark;
    do {
        cout << "Enter Your Mark: ";
        cin >> mark;
        if (mark < 0 || mark > 100) {
            cout << "Invalid Mark, Re-enter (0 - 100)\n";
        }
    } while (mark < 0 || mark > 100);
    cout << "-------------------\nResult:\n";
    pass_or_fail(mark);
    return 0;
}
```

Output

Microsoft Visual Studio Debug Console

```
Enter Your Mark: 69
-------------------
Result:
Passed: D
```

Q2

Source Code

Writing the same program but for 100 students, save their marks in an array, then count how many students passed and how many failed in that course.

assignment_3.cpp

assignment_3

```cpp
#include <iostream>
using namespace std;
#define SIZE 100

int passed = 0;
int failed = 0;

void pass_or_fail(float mark) {
    if (mark >= 50) {
        passed++;
        cout << "Passed: ";
    }
    else {
        failed++;
        cout << "Failed: ";
    }
    if (mark >= 90) {
        cout << "A\n";
    }
    else if (mark >= 80) {
        cout << "B\n";
    }
    else if (mark >= 70) {
        cout << "C\n";
    }
    else if (mark >= 50) {
        cout << "D\n";
    }
    else {
        cout << "F\n";
    }
}
```

```cpp
int main() {
    float marks[SIZE];
    bool valid;

        for (int i = 0; i < SIZE; i++) {
            do {
            cout << "Student " << i + 1 << " Mark: ";
            cin >> marks[i];
            valid = true;
            if (marks[i] < 0 || marks[i] > 100) {
                cout << "Invalid Mark, Re-enter (0 - 100)\n";
                valid = false;
            }
            else {
                pass_or_fail(marks[i]);
            }
        } while (!valid);
        }
        cout << "-------------\nStatistics:\n";
        cout << "Students passed: " << passed << '\n';
        cout << "Students failed: " << failed << '\n';
    return 0;
}
```

Q2

Output

Select Microsoft Visual Studio Debug Console

```
Student 93 Mark: 78
Passed: C
Student 94 Mark: 7
Failed: F
Student 95 Mark: 6
Failed: F
Student 96 Mark: 5
Failed: F
Student 97 Mark: 5
Failed: F
Student 98 Mark: 54
Passed: D
Student 99 Mark: 65
Passed: D
Student 100 Mark: 76
Passed: C
--------------
Statistics:
Students passed: 55
Students failed: 45
```

Q3

What is the main importance of pointers?

Pointers are implemented to modify, retrieve or manipulate variables via their memory address.

Q4

What are dangling pointers?

Dangling pointers are pointing to a freed variable or an out of scope variable, and they can cause unexpected behaviour.

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THANK YOU

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By Dr. Dina Mohamed